

Engineering Standards and Release Runbook

Sao Mai Technology JSC · Engineering · Effective 1 January 2026

These standards apply to every service and application built or maintained by Sao Mai Technology engineering teams. They are enforced by the continuous integration pipeline where possible and by code review where not.

1. Source control

Branch naming	feature/<ticket-id>-<short-description>, for example feature/SM-1420-invoice-export
Direct commits	Direct commits to main are blocked. All changes arrive through a pull request.
Approvals	Two approvals are required. One approval is sufficient where the change is under 50 lines.
Stale branches	Feature branches with no activity for 30 days are deleted automatically.
Commit messages	Imperative mood, referencing the ticket id in the first line.

2. Quality gates

- **Test coverage on new code must be at least 70%.** The build fails below this threshold; it is a gate, not a warning.
- Static analysis runs on every pull request. New critical or blocker issues fail the build.
- A secret scan runs nightly across the whole repository and on every pull request.
- Dependency updates are reviewed monthly. A **critical CVE must be patched within 72 hours** of being published.

3. Release windows

Standard windows	Tuesday and Thursday, 14:00 to 16:00 ICT
Never deploy	On a Friday, on the day before a public holiday, or at any point during the 9-day Tet closure
Emergency releases	Permitted outside these windows for a SEV1 incident only, with the on-call engineering manager's approval
Rollback SLA	A failed release must be rolled back within 15 minutes of detection
Release notes	Published to the internal channel before the window opens

The Tet restriction exists because the on-call roster runs at reduced strength during the closure. See the Employee Handbook for the closure length and the company days that apply.

4. On-call

Rotation length	One week, handed over on Monday at 10:00 ICT
Frequency limit	No engineer may be rostered more than once in any six-week period
Allowance	800,000 VND per week on call, paid with the following month's salary

Time off in lieu	An engineer called out between 22:00 and 06:00 takes the following morning off, with no deduction from annual leave
Escalation	On-call engineer, then team lead, then engineering manager, at 15-minute intervals

5. Incident severity

SEV1	Full outage, or customer data at risk. Response within 15 minutes. Status updates every hour until resolved.
SEV2	Degraded service affecting many customers. Response within 1 hour.
SEV3	Limited or cosmetic impact. Handled in the next working day's triage.
Post-incident review	Required within 5 working days for every SEV1. Blameless format, with actions assigned to named owners.

6. Secrets and data handling

- Secrets are never committed to source control. Any secret found in history is treated as compromised and rotated immediately.
- All credentials are **rotated every 90 days**.
- Production database access requires a ticket and is granted for a maximum of **8 hours**.
- **Data residency:** data belonging to Vietnamese government clients is stored in the Hanoi region and must not be replicated outside Vietnam.
- Customer data is never copied into a development or test environment. Test data is generated, not cloned.

7. Environments

Development	Per-engineer, reset nightly
Staging	Mirrors production configuration. All releases pass through staging first.
Production	Change only through the release pipeline. No manual configuration edits.

Questions about these standards should go to the Engineering Management team. This document is reviewed each January and was last updated on 1 January 2026.